INHA UNIVERSITY IN TASHKENT

Computer Algorithms (CIE3090)

SAMPLE SOLUTION for reference

Q1. Convert the following sequence to a Heap using Build Max Heap

12, 9, 7, 2, 15, 32

Use this space for showing iterations For I → 2 to 0		0	1	2	3	4	5		
		12	**9**	**7**	**2**	**15**	**32**		
i=2 L=5 R=6	**i=2**			**i**			**L**	**Non leaf nodes**	**(n/2)-1 to 0**
		12	**9**	**32**	**2**	**15**	**7**		
i=1 L=3 R=4	**i=1**		**i**		**L**	**R**			
		12	**15**	**32**	**2**	**9**	**7**	**i=3 exit**	
i=0 L=1 R=2	**i=0**	**i**	**L**	**R**					
		32	**15**	**12**	**2**	**9**	**7**	**i=2 L=5 R=6**	**exit**
				i			**L**		
		32	**15**	**12**	**2**	**9**	**7**		

Q2. **Apply Quicksort Algorithm to sort the given sequence**

(Show only steps of first PARTITION on the initial array)

Use this space for any calculations	1	2	3	4	5	6
P=1 r=6 , x=10 initially J=0 I=-1	**9**	**7**	**2**	**15**	**32**	**10**
	A[j]< x i = j					
		A[j]< x i = j				
			A[j]< x i = j	j	j	
				i		x
	9	**7**	**2**	**10**	**32**	**15**

Q2. **Apply Merge sort Algorithm to sort the given sequence**
(Show all steps)

Use this space for any calculations		1	2	3	4	5	6
		9	**7**	**2**	**15**	**32**	**10**
p=1	**r=6**	**9**	**7**	**2**			
q=3		**9**	**7**		**15**	**32**	**10**
				2	**15**	**32**	
		9					**10**
			7		**15**		
		7	**9**			**32**	
		2	**7**	**9**	**15**	**32**	
					10	**15**	**32**
		2	**7**	**9**	**10**	**15**	**32**